

◆ Section 16 — Independence and Safe Decision Gating

16.0 Core Principle

No execution without sufficient independence, and stability must never be violated.

16.0.1 Effective Independence

$$I_{\mathrm{eff}} = 1 - \big(\mathrm{Corr}_{\mathrm{obs}} + \mathrm{Corr}_{\mathrm{hidden}} \big), \\ \quad I_{\mathrm{eff}} \in [0,1]$$

16.0.2 Decision Gate with Hysteresis

$$\gamma = \begin{cases} 1 & \text{if } I_{\mathrm{eff}} \geq \theta_{\mathrm{on}} \\ 0 & \text{if } I_{\mathrm{eff}} \leq \theta_{\mathrm{off}} \end{cases}, \\ \quad \theta_{\mathrm{on}} >$$

θ_{off}

16.0.3 System Dynamics

$$\dot{x} = F(x, t) + \gamma u(t)$$

16.0.4 Control Influence Bound

$$\begin{aligned} |\nabla V \cdot u| &\leq B, \\ \text{quad } B &= \|\nabla V\|, u_{\max} \end{aligned}$$

16.0.5 Stability Condition

$$\begin{aligned} \dot{V} &= \nabla V \cdot F + \gamma \nabla V \cdot u \\ &\leq 0 \end{aligned}$$

Assume:

$$\nabla V \cdot F \leq -\alpha \|x\|^2$$

Then it suffices that:

$$\nabla V \cdot u \leq \alpha \|x\|^2$$

16.0.6 Safe Control Set

$$\begin{aligned} \gamma = 1 &\Rightarrow \\ u(t) &\in \mathcal{U}_{\mathrm{safe}} \\ &= \\ \{u: \nabla V \cdot u &\leq \alpha \|x\|^2\} \end{aligned}$$

Thus:

independence enables execution, while the safe set enforces stability.

16.0.7 Independence Threshold

$$\begin{aligned} I_{\mathrm{eff}} &\geq \theta_{\mathrm{ind}} \\ &\Rightarrow \\ u &\in \mathcal{U}_{\mathrm{safe}} \end{aligned}$$

16.0.8 Operational Rules

- $I_{\mathrm{eff}} < \theta_{\mathrm{off}} \Rightarrow \gamma = 0$
- Execution is always restricted to

$\mathcal{U}_{\mathrm{safe}}$

- Increasing correlation $\Rightarrow I_{\mathrm{eff}}$

$\Downarrow \Rightarrow$ gate closes

- Loss of confidence $\Rightarrow$ suppression, not escalation

Core Insight

Independence determines whether to act.

Lyapunov constraints determine how to act safely.

16.1 Structural Independence

Definition

$C_{\mathrm{struct}} =$

$\max_{i \neq j}$

$\frac{|\mathcal{S}_i \cap \mathcal{S}_j|}{|\mathcal{S}_i \cup \mathcal{S}_j|}$

where:

$\mathcal{S}_i = \{\text{source, model, pathway, timing}\}$

Integration

$$I_{\mathrm{eff}} = 1 - \frac{C_{\mathrm{obs}} + C_{\mathrm{struct}} + C_{\mathrm{other}}}{C_{\mathrm{obs}} + C_{\mathrm{struct}} + C_{\mathrm{other}}}$$

Acceptance Condition

$$C_{\mathrm{struct}} \leq \epsilon_{\mathrm{struct}}$$

Implication

$$C_{\mathrm{struct}} \uparrow \Rightarrow I_{\mathrm{eff}} \downarrow \Rightarrow \gamma = 0$$

Principle

Different outputs without different structure = false independence.

16.2 Time Consistency

Temporal Weight

$$w_{\text{time}}^{\{ij\}} = \exp\left(-\frac{|\Delta t_{ij}|}{\tau}\right)$$

Aggregate Measure

$$C_{\text{time}} = 1 - \min_{i \neq j} w_{\text{time}}^{\{ij\}}$$

Constraint

$$|\Delta t_{ij}| \leq \tau_{\text{max}}$$

Integration

$$I_{\mathrm{eff}} = \frac{1 - \big(\mathrm{Corr}_{\mathrm{obs}} + C_{\mathrm{struct}} + C_{\mathrm{time}} + C_{\mathrm{other}}\big)}{2}$$

Principle

True triangulation occurs within a shared temporal window.

16.3 Anti-Amplification (Feedback Suppression)

Loop Detection

$$C_{\mathrm{loop}} \in [0,1]$$

or binary:

$$C_{\mathrm{loop}} = \begin{cases} 1 & \text{feedback detected} \\ 0 & \text{otherwise} \end{cases}$$

Integration

$$I_{\mathrm{eff}} = 1 - \big(\dots + C_{\mathrm{loop}} \big)$$

Implication

$$C_{\mathrm{loop}} \uparrow \rightarrow I_{\mathrm{eff}} \downarrow \rightarrow \gamma = 0$$

Principle

The system must never trust the same signal twice.

16.4 Conservative Mode

Activation

$I_{\mathrm{eff}} < \theta_{\mathrm{ind}}$

\\;\text{or}\\;

$C_{\mathrm{time}} > \tau_{\mathrm{max}}$

\\;\text{or}\\;

$C_{\mathrm{struct}} > \epsilon_{\mathrm{struct}}$

Behavior

$\gamma = 0$

Properties

- No execution
- No escalation
- Re-evaluation only

Final Principle

The system prefers silence over error.

SEKHEM-TSDD Theory — Mohamed Abd Elnaby
DOI: 10.5281/zenodo.19521219